

TUGAS JURNAL-03

MODUL 9

2311104004

Marsella dwi julianti

SE07-01

Link Github:

https://github.com/marsellacell/KPL_MARSELLADWIJULIANTI_2311104004_SE0701/tree/marsella/09_API-Design-dan-Construction-Using-Swagger/JURNAL-03

Source dan penjelasan:



```
1 const express = require('express');
2 const swaggerUi = require('swagger-ui-express');
3 const swaggerJsDoc = require('swagger-jsdoc');
```

Code diatas untuk mengimport library pakai express untuk server API, swagger untuk dokumentasi.



```
1 const app = express();
2 const PORT = 3000;
3
4 // Middleware
5 app.use(express.json());
```

Code diatas untuk bikin server dan supaya bisa baca data JSON dari body request.



```
1 let mahasiswa = [
```

Code diatas untuk data 3 Mahasiswa langsung disimpan di array, tanpa database.



```
1 app.get('/api/Mahasiswa', (req, res) => {
```

Code diatas untuk endpoint get semua mahasiswa dan mengambil semua data Mahasiswa.



```
1 app.get('/api/Mahasiswa/:id', (req, res) => {
```

Code diatas untuk endpoint get mahasiswa by ide dan mengambil 1 Mahasiswa berdasarkan index ID.



```
1 app.post('/api/Mahasiswa', (req, res) => {
```

Code diatas untuk endpoint tambah mahasiswa atau menambahkan data Mahasiswa baru.



```
1 app.delete('/api/Mahasiswa/:id', (req, res) => {
```

Code diatas untuk endpoint delete mahasiswa by ide atau menghapus data Mahasiswa berdasarkan ID.

```

1  const swaggerOptions = {
2    swaggerDefinition: {
3      openapi: '3.0.0',
4      info: {
5        title: 'Mahasiswa API',
6        version: '1.0.0',
7        description: 'API for managing Mahasiswa data',
8      },
9    },
10   apis: ['./server.js'], // path ke file ini
11 };
12
13 const swaggerDocs = swaggerJSDoc(swaggerOptions);
14 app.use('/swagger', swaggerUi.serve, swaggerUi.setup(swaggerDocs));

```

Code diatas Untuk generate halaman dokumentasi API di <http://localhost:3000/swagger>.

```

1  app.listen(PORT, () => {
2    console.log(`Server running at http://localhost:${PORT}`);
3  });

```

Code diatas untuk memulai server di port 3000.

Output:

GET /api/Mahasiswa

The screenshot shows the Swagger UI for the endpoint `GET /api/Mahasiswa`. The response is a 200 status code with a JSON body containing a list of three students. The response headers show `content-length: 286`, `content-type: application/json; charset=utf-8`, `date: Sun, 27 Apr 2025 17:47:22 GMT`, `etag: W/"11e-43j6q4vciH8m17v5f04u8jH5ao"`, and `x-powered-by: Express`.

Code	Description	Links
200	Successful retrieval of Mahasiswa list	No links

POST /api/Mahasiswa

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:3000/api/Mahasiswa' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Bagus Pratama",
    "nim": "2311104055",
    "course": ["KPL", "Pemrograman Web"],
    "year": 2022
  }'
```

Request URL

http://localhost:3000/api/Mahasiswa

Server response

Code	Details
201	Response body Mahasiswa created successfully Response headers <pre>connection: keep-alive content-length: 30 content-type: text/html; charset=utf-8 date: Sun, 27 Apr 2025 17:49:36 GMT etag: W/"1e-Md1GdPlewofguXISQpHhPevJH" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

Code	Description	Links
201	Mahasiswa created successfully	No links

GET /api/Mahasiswa

Server response

Code	Details
200	Response body <pre>{ "Database": "PBO", "year": 2022, { "name": "Ahmad Fadli", "nim": "2311104012", "course": ["Jaringan", "AI", "KPL"], "year": 2022 }, { "name": "Nurul Hidayati", "nim": "2311104021", "course": ["Machine Learning", "Basis Data"], "year": 2022 }, { "name": "Bagus Pratama", "nim": "2311104055", "year": 2022 } }</pre> Response headers <pre>connection: keep-alive content-length: 377 content-type: application/json; charset=utf-8 date: Sun, 27 Apr 2025 17:57:56 GMT etag: W/"179-JuEbx8duYu3zPk7gzfMzAvI+c" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Successful retrieval of Mahasiswa list	No links

GET /api/Mahasiswa/3

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:3000/api/Mahasiswa/3' \
  -H 'accept: */*'
```

Request URL

http://localhost:3000/api/Mahasiswa/3

Server response

Code	Details
200	Response body <pre>{ "name": "Bagus Pratama", "nie": "2311104055", "course": { "KPL", "Pemrograman Web" }, "year": 2022 }</pre> Response headers <pre>content-length: 99 content-type: application/json; charset=utf-8 date: Sun, 27 Apr 2025 17:59:42 GMT etag: W/"5a-1l0Uzm6QKkJTExqS18Yt3yTKA" x-powered-by: Express</pre>

Responses

Code	Description	Links
200	Successful retrieval of a Mahasiswa	No links
404	Mahasiswa not found	No links

DELETE /api/Mahasiswa/1

id * required ID index of Mahasiswa to delete

integer (path)

1

Execute Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:3000/api/Mahasiswa/1' \
  -H 'accept: */*'
```

Request URL

http://localhost:3000/api/Mahasiswa/1

Server response

Code	Details
200	Response body <pre>Mahasiswa deleted successfully</pre> Response headers <pre>connection: keep-alive content-length: 30 content-type: text/html; charset=utf-8 date: Sun, 27 Apr 2025 18:00:46 GMT etag: W/"1a-w1LeQ5RntBpdKu4eTNe91Temlo" keep-alive: timeout=5 x-powered-by: Express</pre>

Responses

GET /api/Mahasiswa

Request URL

http://localhost:3000/api/mahasiswa

Server response

CodeDetails

200

Response body

[{"name": "Marsella Ovi Julianti", "nim": "2311104004", "course": [{"xpl": "Database", "pbo": ""}], "year": 2022}, {"name": "Nurul Hidayati", "nim": "2311104021", "course": [{"xpl": "Machine Learning", "pbo": "Basis Data"}], "year": 2022}, {"name": "Bagus Pratama", "nim": "2311104055", "course": [{"xpl": "Pemrograman Web"}], "year": 2022}]

Explain

Download

Response headers

connection: keep-alive
content-length: 299
content-type: application/json; charset=utf-8
date: Sun, 27 Apr 2025 18:00:59 GMT
etag: W/"122-2IG9HvAYc2VL3ATony8H4317s6E"
keep-alive: timeout=5
x-powered-by: Express

Responses

CodeDescription

Links